

Problem Set 1

Wen Sheng (ws2840)

2026-10-02

Problem 1

(i)

By Definition,

$$\begin{aligned} E[Y|X = x] &= \sum_y y \times p(y|X = x) \\ &= \sum_y y \times \frac{P(X = x, Y = y)}{P(X = x)} \end{aligned}$$

The joint probability distribution $P(X = x, Y = y)$ is:

$X \backslash Y$	2	3	4	5	6	7	8	9	10	11	12	$P(X = x)$
1	$\frac{1}{36}$	0	0	0	0	0	0	0	0	0	0	$\frac{1}{36}$
2	0	$\frac{2}{36}$	$\frac{1}{36}$	0	0	0	0	0	0	0	0	$\frac{3}{36}$
3	0	0	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	0	0	0	0	0	0	$\frac{5}{36}$
4	0	0	0	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	0	0	0	0	$\frac{7}{36}$
5	0	0	0	0	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	0	0	$\frac{9}{36}$
6	0	0	0	0	0	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{11}{36}$
$P(Y = y)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	1

So,

$$E[Y|X = 1] = 2 * \frac{\frac{1}{36}}{\frac{1}{36}} = 2$$

$$E[Y|X = 2] = 3 * \frac{\frac{2}{36}}{\frac{2}{36}} + 4 * \frac{\frac{1}{36}}{\frac{2}{36}} = \frac{10}{3}$$

$$E[Y|X = 3] = 4 * \frac{\frac{2}{36}}{\frac{5}{36}} + 5 * \frac{\frac{2}{36}}{\frac{5}{36}} + 6 * \frac{\frac{1}{36}}{\frac{5}{36}} = \frac{24}{5}$$

$$E[Y|X = 4] = 5 * \frac{\frac{2}{36}}{\frac{7}{36}} + 6 * \frac{\frac{2}{36}}{\frac{7}{36}} + 7 * \frac{\frac{2}{36}}{\frac{7}{36}} + 8 * \frac{\frac{1}{36}}{\frac{7}{36}} = \frac{44}{7}$$

$$E[Y|X = 5] = 6 * \frac{\frac{2}{36}}{\frac{9}{36}} + 7 * \frac{\frac{2}{36}}{\frac{9}{36}} + 8 * \frac{\frac{2}{36}}{\frac{9}{36}} + 9 * \frac{\frac{2}{36}}{\frac{9}{36}} + 10 * \frac{\frac{1}{36}}{\frac{9}{36}} = \frac{70}{9}$$

$$E[Y|X = 6] = 7 * \frac{\frac{2}{36}}{\frac{11}{36}} + 8 * \frac{\frac{2}{36}}{\frac{11}{36}} + 9 * \frac{\frac{2}{36}}{\frac{11}{36}} + 10 * \frac{\frac{2}{36}}{\frac{11}{36}} + 11 * \frac{\frac{2}{36}}{\frac{11}{36}} + 12 * \frac{\frac{1}{36}}{\frac{11}{36}} = \frac{102}{11}$$

(ii)

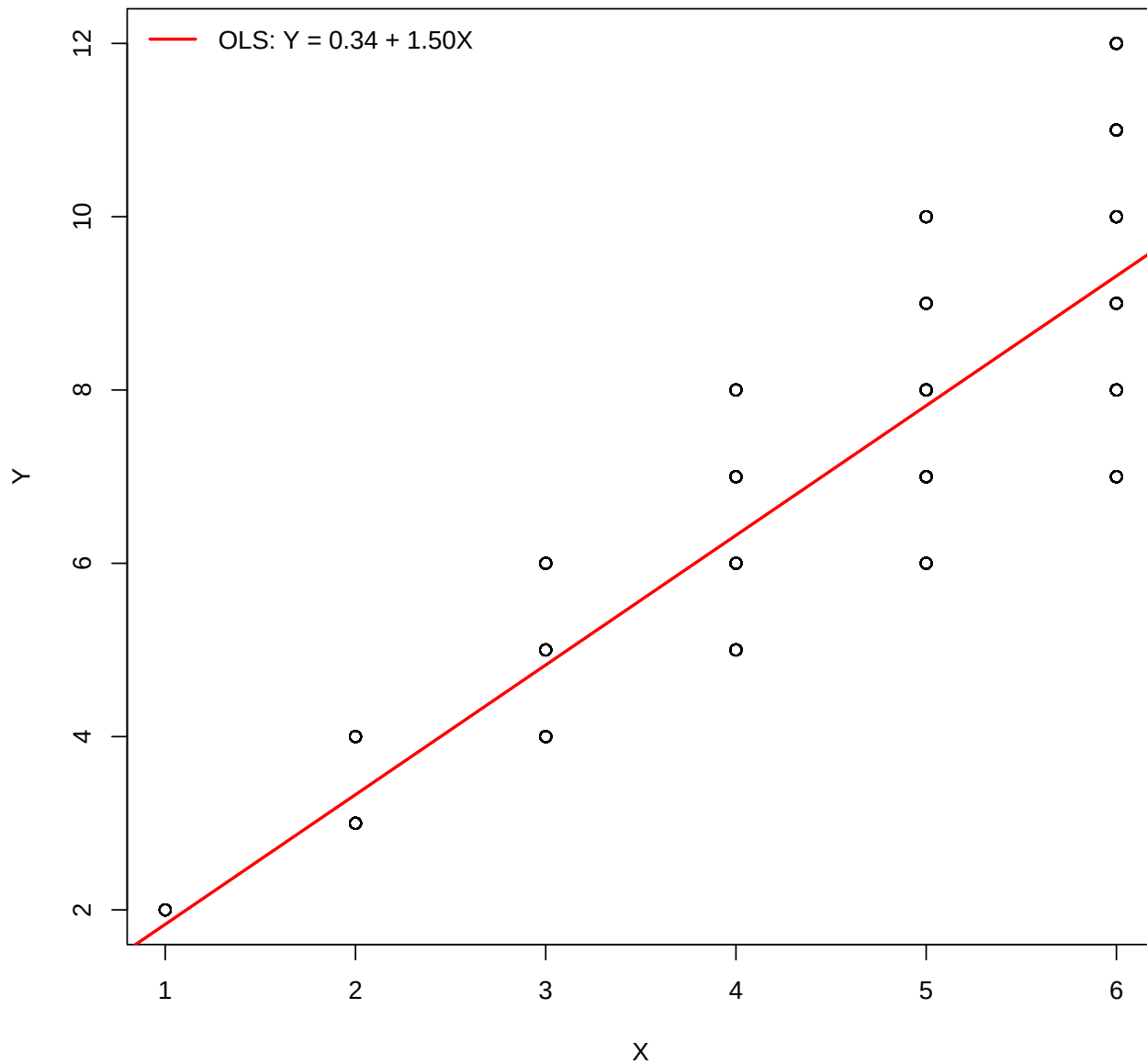
Simulate $n = 1000$ draws of (X, Y) . Plot the draws:

```
set.seed(17)
n <- 1000
D1 <- sample(1:6, n, replace = TRUE)
D2 <- sample(1:6, n, replace = TRUE)

X <- pmax(D1, D2)
Y <- D1 + D2
ols <- lm(Y ~ X)

plot(X, Y, main = "Sum and Max of Two Dice Rolls")
abline(ols, col = "red", lwd = 2)
legend("topleft",
      sprintf("OLS: Y = %.2f + %.2fX", coef(ols)[1], coef(ols)[2]),
      col = "red", lwd = 2, bty = "n")
```

Sum and Max of Two Dice Rolls



(iii)

After simulation $n = 1000$ observations and fitting the OLS regression,

$$\hat{\beta}_0 = 0.34, \quad \hat{\beta}_1 = 1.50$$

so

$$E[Y|\widehat{X} = x] = \hat{\beta}_0 + \hat{\beta}_1 x = 0.34 + 1.50x$$

(iv)

(v)

Problem 2

Problem 3

(i)

The log-likelihood for T i.i.d. observations is the following. We can prove the result.

(ii)

(iii)

The log-likelihood for T i.i.d. observations is the following. We can prove the result.

(iv)

Problem 4

(i)

The log-likelihood for T i.i.d. observations is the following. We can prove the result.

(ii)

Problem 5

(i)

The log-likelihood for T i.i.d. observations is the following. We can prove the result.

(ii)

Problem 6

(i)

The log-likelihood for T i.i.d. observations is the following. We can prove the result.

(ii)

(iii)

```
library(ggplot2)

ggplot(airquality, aes(Temp, Ozone)) +
  geom_point() +
  geom_smooth(method = "loess")
```

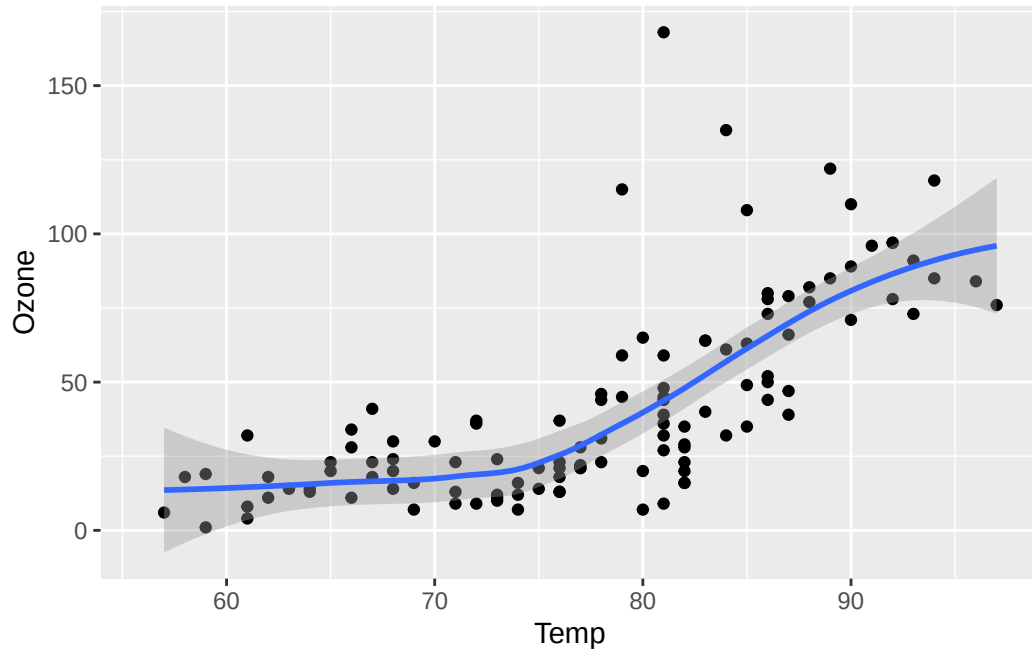


Figure 1: Temperature and ozone level.